

Literature Review

1st MingYi Wei

Department of Electrical and Computer Engineering
University of Illinois Urbana-Champaign

Urbana, USA
mingyi5@illinois.edu

Abstract—This literature review assignment is focus on my final project topic: Precision Biological Illustration Generation via Multi-Agent Collaboration and Knowledge Base. The following is the Abstract and Introduction of my final project main topic. Generative Diffusion Models often suffer from "hallucinations," producing anatomically incorrect biological illustrations. This project proposes BioGuard-Diffusion, a framework leveraging Multi-Agent systems and Retrieval-Augmented Generation (RAG) grounded in the AI2D dataset [1] to ensure structural and nomenclature precision in AI-generated scientific images.

Index Terms—Generative AI, Diffusion Models, Retrieval-Augmented Generation (RAG), Multi-Agent Systems, Biological Illustrations, AI2D Dataset, Scientific Accuracy.

I. INTRODUCTION

Text-to-image diffusion models have shown remarkable synthesis capabilities, such as Gemini, ChatGPT, but often fail in domain-specific tasks requiring high structural integrity. In biological sciences, "hallucinations", such as misplacing a protein receptor or mislabeling a bone name, as shown in figure 1, let AI-generated images educationally invalid. This project addresses these limitations by introducing BioGuard-Diffusion, a "Reasoning-before-Generation" pipeline. Our approach builds upon the importance of semantic parsing in scientific images, questions and annotations established by the AI2D dataset and the potential of Retrieval-Augmented Generation [2] [3] to improve synthesis. By leveraging Multi-Agent [4] Collaboration frameworks, we decompose each steps into specialized agents, significantly reducing hallucinations compared to single-prompt models and ensuring the anatomical accuracy of generated biological illustrations.

II. RELATED WORK

A. Text-to-Image Diffusion Models in Scientific Domains

Recent advancements in Text-to-Image diffusion models, including DALL-E 3, Midjourney v6.0, and Stable

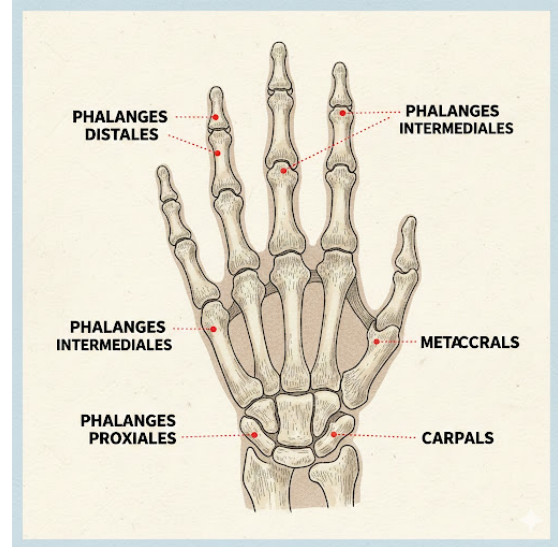


Fig. 1. Example of mislabeling images generated from Gemini

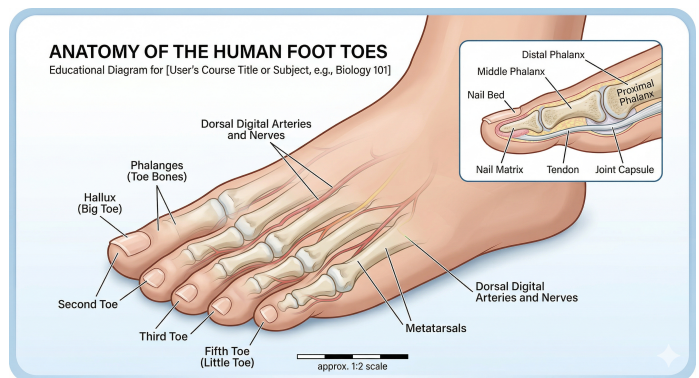


Fig. 2. Generate examples of error indicators and unclear markings

Diffusion, have demonstrated remarkable synthesis capabilities for general-purpose imagery. These models have gained significant attention for their potential to automate the production of educational and medical materials due to their stylistic versatility and rapid generation speeds. However, quantitative evaluations indicate a persistent

trade-off between aesthetic quality and factual accuracy. While Midjourney v6.0 often scores highest in aesthetic appeal, it frequently fails to meet the rigorous standards of anatomical detail required for medical education.

The primary obstacle to adopting these models in scientific domains is the phenomenon of "image hallucination" [11]. In the context of scientific imaging, hallucinations are defined as AI-fabricated abnormalities [12] or artifacts that appear visually realistic but are factually false and deviate from anatomical or functional truth. These inaccuracies render AI-generated illustrations educationally invalid and potentially dangerous if deployed in clinical settings, where they may lead to misdiagnosis or patient misinformation.

Systematic comparative studies have documented consistent failures across various anatomical structures:

Skeletal Systems: Research [13] evaluating the human skull and ribs found that models frequently omit crucial landmarks, such as the mental and supraorbital foramina. In rib cage generation, models consistently fail to represent the correct number of ribs, often omitting the 11th and 12th floating ribs and mislabeling the articulation of false ribs.

Neuroanatomy [14]: Illustrations of the human brain often lack accurate depictions of the gyri and sulci. Furthermore, the spatial relationship between the cerebellum and temporal lobes remains unclear and poorly illustrated in single-prompt generations.

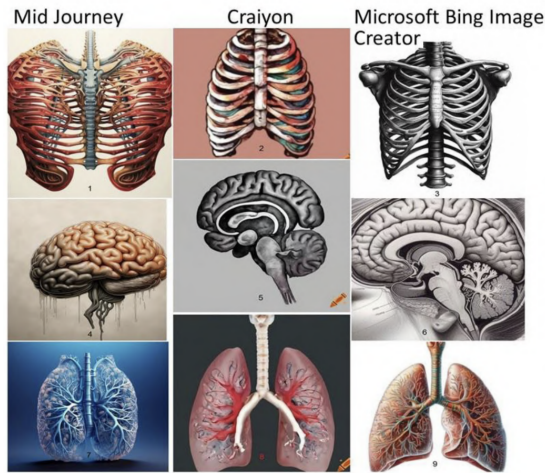


Fig. 3. Example of single-prompt generations result from different model

Craniofacial and Cardiovascular Structures [15]: Models struggle with complex 3D relationships, resulting in "mixing of layers" where superficial skin is juxtaposed with deep skeletal structures. In cardiovascular imaging,

models fail to indicate proper coronary artery origins, and the branching patterns of the aorta and pulmonary trunk are consistently incorrect.

These limitations [16] stem from "co-occurrence bias," where models rely on dominant patterns in training data—such as the modern appearance of an object—at the expense of specific historical or scientific facts requested in a prompt. Furthermore, most current models prioritize visual image quality (e.g., using mean squared error loss) rather than the preservation of structural integrity. Consequently, while GAI models offer a promising frontier for rapid prototyping, they currently require significant expert refinement to approach medical education standards

B. Knowledge-Grounded Image Generation (RAG-Diffusion)

Standard text-to-image (T2I) diffusion models are fundamentally constrained by their internal knowledge, which is fixed within model parameters during training on closed datasets. This limitation often leads to significant "hallucinations" or structural distortions when the model encounters fine-grained or domain-specific objects that were underrepresented in the training phase. To bridge this knowledge gap, Retrieval-Augmented Generation (RAG) has emerged as a pivotal technique, transitioning from language tasks to the vision domain to mitigate hallucinations by integrating external, verifiable knowledge sources.

Recent frameworks like RealRAG [17] have addressed the "unseen object" problem by proposing a real-object-based retrieval-augmented framework. By employing self-reflective contrastive learning, RealRAG retrieves high-quality real-world images to compensate for the generative model's missing memory, ensuring that fine-grained details, similar to the specific biological structures in this project, remain realistic and accurate. In the scientific domain, specialized systems such as eNR-RCrew [18] [19] have demonstrated the power of combining multi-agent collaboration with GraphRAG. This approach allows for the extraction of reliable structure-activity relationships from vast, heterogeneous literature, providing a "reasoning-before-generation" foundation that overcomes the lack of deep domain reasoning in generic LLMs.

The mechanism of "reference-guided" generation is further supported by the RATD [20] framework. While initially designed for time series, its core logic of retrieving meaningful samples from a database to guide the denoising process is highly relevant to biological

illustration. This reference-guided mechanism ensures that the diffusion process is not merely a probabilistic guess but is anchored by historical or factual templates. Furthermore, knowledge-augmented workflows in other disciplines, such as architectural massing generation, have successfully utilized prototype libraries and Knowledge Graph (KG) routing [21] to map complex narratives into executable strategy tokens. This project builds upon these concepts by implementing a Biologist Agent that queries a biological knowledge base, transforming retrieved factual constraints (e.g., organelle morphology) into structural anchors for the generation layer

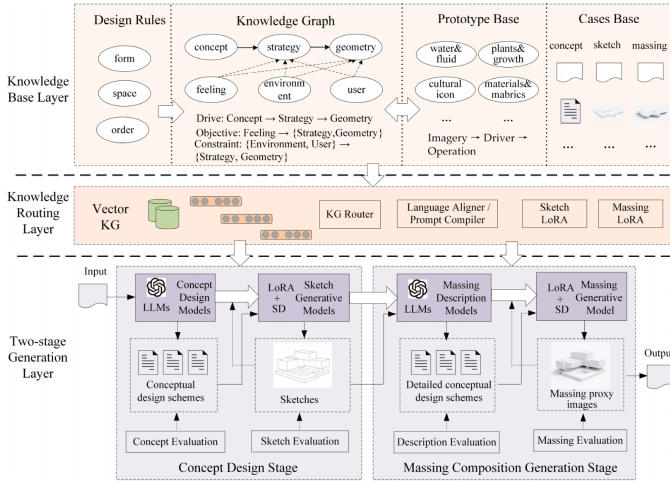


Fig. 4. A knowledge-augmented two-stage workflow for architectural concept-to-massing generation and evaluation.

C. Multi-Agent Systems for Generative AI

Recent advancements in text-to-image generation have increasingly shifted from monolithic single-prompt inference toward Multi-Agent Systems (MAS) to address complex compositional and domain-specific challenges. The foundational strategy of separating high-level logical planning from visual execution was exemplified by frameworks such as LayoutGPT [22], which utilizes Large Language Models (LLMs) as visual planners to generate precise 2D layouts before image synthesis, significantly enhancing spatial and numerical reasoning compared to direct generation. Building on this modular approach, MCCD [23] introduced a multi-agent collaboration-based scene parsing module that decomposes complex prompts into a specialized agent system, where individual agents manage specific objects and their inter-relations to overcome performance bottlenecks in complex scene generation.

The necessity for rigorous quality control and iterative correction in generative workflows is addressed by frameworks like Mac-Tiger [24]. Mac-Tiger employs a tri-agent architecture, comprising a Reviewer, Challenger, and Refiner, to leverage multimodal large language models (MLLMs) for iterative optimization, ensuring that the final output maintains high fidelity and alignment with complex instructions. The structural efficiency of such systems is further supported by recent research into Scaling Agent Systems, which provides quantitative evidence that centralized coordination structures are superior in structured tasks for reducing error diffusion compared to independent or decentralized agent groups. Furthermore, the application of MAS in high-precision scientific domains is demonstrated by CellAgent [25], an LLM-driven multi-agent framework for automated single-cell data analysis. By decomposing labor-intensive biological processing tasks into specialized agent roles, CellAgent proves the viability and effectiveness of multi-agent collaboration in maintaining accuracy within complex scientific environments.

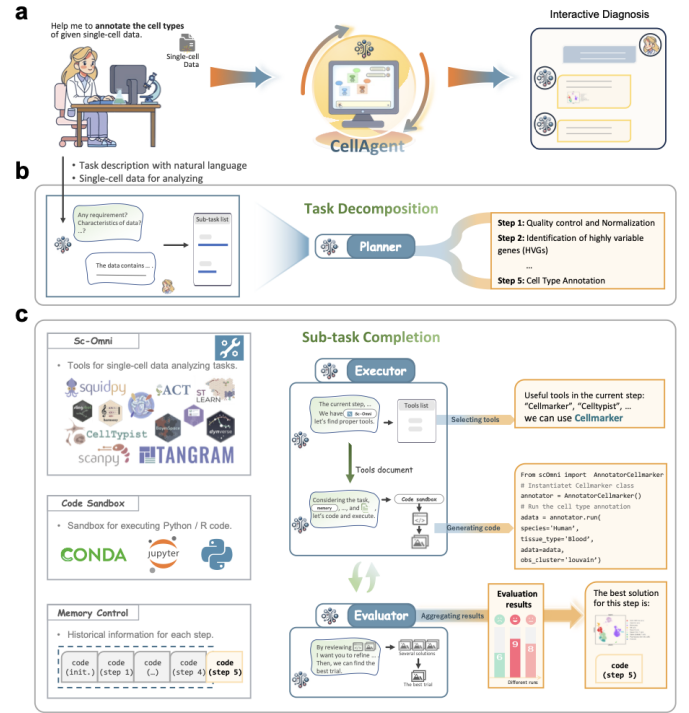


Fig. 5. Schematic of the CellAgent Framework

D. Evaluation Metrics for Biological Accuracy

Standard evaluation metrics for text-to-image (T2I) generation, such as Fréchet Inception Distance (FID) and the general-purpose CLIP-Score, primarily assess

visual fidelity and coarse-grained semantic alignment. However, these metrics often fail to capture the high structural integrity and anatomical precision required for biological and medical illustrations, where a minor visual hallucination can render a scientific image invalid. To address this, recent research has shifted toward domain-specific foundation models and structure-aware evaluation frameworks.

The foundation for biology-specific vision-language alignment was established by BioCLIP [26], which leverages the hierarchical "Tree of Life" taxonomy to significantly outperform general-purpose models like OpenCLIP on biological classification and retrieval tasks. This capability was further refined in BioCLIP 2 [27], which introduces hierarchical contrastive learning at scale. BioCLIP 2 demonstrates emergent properties where embeddings naturally align with ecological and morphological relationships, such as species-specific traits and intra-species variations. This provides a robust backbone for our Verifier Agent to perform fine-grained anatomical cross-checking.

Moreover, the challenge of cross-modal accuracy in medical contexts is addressed by BioMedBLIP [28], which utilizes bootstrapped language-image pretraining to improve performance on medical VQA and image captioning tasks. This framework offers a methodology for validating whether a generated illustration accurately reflects the scientific constraints described in the text. To refine the scoring of these complex structures, research into Hyperbolic Multimodal Representation Learning [29] suggests that hyperbolic embedding spaces are intrinsically superior to Euclidean spaces for modeling the parent-child relationships inherent in biological taxonomies. By utilizing hyperbolic distance, we can better quantify "anatomical distance," ensuring that our Scientific Accuracy Score (SAS) [30] weighs errors based on their taxonomic or structural significance—for instance, penalizing the misplacement of a major organelle more heavily than minor stylistic variations.

REFERENCES

- [1] Kembhavi, Aniruddha, Michael Salvato, Eric Kolve, Minjoon Seo, Hannaneh Hajishirzi and Ali Farhadi. "A Diagram is Worth a Dozen Images." ArXiv abs/1603.07396 (2016): n. pag.
- [2] Ge, Yuying, Sijie Zhao, Wang Zhao, Yixiao Ge, Wei Liu and Ying Shan. "SEED-X: Multimodal Models with Unified Multi-granularity Comprehension and Generation." Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) (2024).
- [3] Vorster, Casper, Joeri Hermans and Gilles Louppe. "Symphony: Learning Expressive Text-to-Image Layout Rectification." ArXiv abs/2412.09614 (2024): n. pag.
- [4] Lian, Long, Boyi Li, Adam Yala and Trevor Darrell. "LLM-grounded Diffusion: Enhancing Layout-to-Image Generation with Large Language Models." Proceedings of the Annual Conference on Neural Information Processing Systems (NeurIPS) (2024).
- [5] Zheng, Guangcong, Jiaxin Ma, Jidong Ji, Haoran Tang and Li Aiquan. "LayoutDiffusion: Improving Graphic Layout Generation by Discrete Diffusion Probabilistic Models." Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) (2024).
- [6] Kembhavi, Aniruddha, Mike Salvato, Eric Kolve, Minjoon Seo, Hannaneh Hajishirzi and Ali Farhadi. "A Corpus of Primary School Science Diagrams." Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR) (2016).
- [7] Stevens, Samuel, Jiaman Wu, Matthew J. Thompson, Elizabeth G. Campolongo, Chan Hee Song, David Edward Alexander Chagaris, Pau Seguí, Bryan Heidorn and Chao-Yuan Wu. "BioCLIP: A Multimodal Foundation Model for Biodiversity." Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) (2024).
- [8] Zhang, Sheng, Yanbo Xu, Naoto Usuyama, Jaspreet Bagga, Robert Tinn, Sam Preston, Rajesh Rao, Christian S. Lorentzen, Mu Wei, Cliff Wong, Matthew P. Lungren, Tristan Naumann and Hoifung Poon. "Large-Scale Domain-Specific Pretraining for Biomedical Vision-Language Processing." arXiv preprint arXiv:2303.00915 (2023).
- [9] Wang, Zhibo, Sihan Chen and Jing Liu. "SciDiff: Generating Scientific Diagrams via Layout-conditioned Diffusion Models." Proceedings of the ACM International Conference on Multimedia (2024).
- [10] Sun, S.; Fu, Y. A Knowledge-Augmented Two-Stage Workflow for Architectural Concept-to-Massing Generation and Evaluation. *Buildings* 2026, 16, 1265. <https://doi.org/10.3390/buildings16061265>
- [11] Lim, Y., Choi, H., Shim, H. (2025). Evaluating Image Hallucination in Text-to-Image Generation with Question-Answering. *Proceedings of the AAAI Conference on Artificial Intelligence*, 39(25), 26290-26298. <https://doi.org/10.1609/aaai.v39i25.34827>
- [12] Xia, Menghua et al. "On Hallucinations in Artificial Intelligence-Generated Content for Nuclear Medicine Imaging (the DREAM Report)." *Journal of Nuclear Medicine* 67 (2025): 166 - 174.
- [13] Noel GPJC. Evaluating AI-powered text-to-image generators for anatomical illustration: A comparative study. *Anat Sci Educ.* 2024;17:979–983. <https://doi.org/10.1002/ase.2336>
- [14] Sikandar, Ayesha Bahar, Hafiza. (2024). A comparative analysis to assess anatomical illustrations via three AI-Driven Text-to-Image generators. *European Journal of Anatomy*. 28. 631-634. 10.52083/DLQJ8889.
- [15] Haider, S.A.; Prabha, S.; Gomez-Cabello, C.A.; Borna, S.; Pressman, S.M.; Genovese, A.; Trabilsy, M.; Galvao, A.; Aziz, K.T.; Murray, P.M.; et al. A Validity Analysis of Text-to-Image Generative Artificial Intelligence Models for Craniofacial Anatomy Illustration. *J. Clin. Med.* 2025, 14, 2136. <https://doi.org/10.3390/jcm14072136>
- [16] Lim, Y., Shim, H. (2024). Addressing Image Hallucination in Text-to-Image Generation through Factual Image Retrieval. ArXiv, abs/2407.10683.
- [17] Lyu, Yuanhuiyi et al. "RealRAG: Retrieval-augmented Realistic Image Generation via Self-reflective Contrastive Learning." ArXiv abs/2502.00848 (2025): n. pag.

- [18] Hao Li, AI-driven multi-agent collaborations for accelerating catalyst design, *National Science Review*, Volume 13, Issue 3, February 2026, nwag067, <https://doi.org/10.1093/nsr/nwag067>
- [19] Xu Hu, Suyu Chen, Letian Chen, Huijuan Wang, Xu Zhang, Zhen Zhou, Automating structure–activity analysis for electrochemical nitrogen reduction catalyst design through multi-agent collaborations, *National Science Review*, Volume 12, Issue 11, November 2025, nwaf372, <https://doi.org/10.1093/nsr/nwaf372>
- [20] Liu, Jingwei et al. “Retrieval-Augmented Diffusion Models for Time Series Forecasting.” *ArXiv abs/2410.18712* (2024): n. pag.
- [21] MDPI and ACS Style Sun, S.; Fu, Y. A Knowledge-Augmented Two-Stage Workflow for Architectural Concept-to-Massing Generation and Evaluation. *Buildings* 2026, 16, 1265.
- [22] Feng, Weixi et al. “LayoutGPT: Compositional Visual Planning and Generation with Large Language Models.” *ArXiv abs/2305.15393* (2023): n. pag.
- [23] Li, Mingcheng et al. “MCCD: Multi-Agent Collaboration-based Compositional Diffusion for Complex Text-to-Image Generation.” *2025 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)* (2025): 13263-13272.
- [24] Luo, Yucong, and Mingyue Cheng. “Mac-Tiger: Multi-Agent Cooperation for Enhanced Text-to-Image Generation.” *Open-Review*, 2026, openreview.net/forum?id=LHysuZBJSu.
- [25] Xiao, Yihang et al. “CellAgent: LLM-Driven Multi-Agent Framework for Natural Language-Based Single-Cell Analysis.” *bioRxiv* (2025): n. pag
- [26] Stevens, Samuel et al. “BioCLIP: A Vision Foundation Model for the Tree of Life.” *2024 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)* (2023): 19412-19424.
- [27] Gu, Jianyang et al. “BioCLIP 2: Emergent Properties from Scaling Hierarchical Contrastive Learning.” *ArXiv abs/2505.23883* (2025): n. pag.
- [28] Naseem, Usman et al. “Advancing Accuracy in Multimodal Medical Tasks Through Bootstrapped Language-Image Pre-training (BioMedBLIP): Performance Evaluation Study.” *JMIR Medical Informatics* 12 (2024): n. pag.
- [29] Gong, ZeMing et al. “Hyperbolic Multimodal Representation Learning for Biological Taxonomies.” *ArXiv abs/2508.16744* (2025): n. pag.
- [30] Chicco, Davide and Giuseppe Jurman. “The advantages of the Matthews correlation coefficient (MCC) over F1 score and accuracy in binary classification evaluation.” *BMC Genomics* 21 (2020): n. pag.